The cogmod R Package: Easy-to-Use Bayesian Cognitive Models for Reaction Times, Choices and Subjective Ratings, with Applications to Computational Neuropsychology

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Cognitive tasks produce structured distributions of reaction times, choices, and subjective ratings, yet these data are often reduced to condition means before analysis. This can obscure differences in variability, distributional shape, and the latent processes that generate behaviour. We introduce `cogmod`, an R package that brings Bayesian cognitive models into the regression workflow used by psychologists. `cogmod` implements descriptive response distributions, evidence accumulation models, and models for bounded ratings as brms custom families, allowing model parameters to be predicted by covariates and random effects using familiar formula syntax. Fitted models remain standard brmsfit objects and therefore support established tools for priors, diagnostics, posterior predictive checks, model comparison, and post-processing. Across worked examples, we show that this framework enables direct comparison of competing decision models, quantifies the reliability of model parameters as individual-level measures, and reveals clinically relevant differences that are invisible in conventional mean-based analyses. By placing models spanning the descriptive-to-generative continuum within a common Bayesian regression framework, `cogmod` lowers the practical barrier to richer analyses of cognitive data and makes the choice of model an empirical question rather than a software constraint. `cogmod` is openly available at https://github.com/DominiqueMakowski/cogmod.

Keywords: cognitive modelling, Bayesian statistics, reaction times, evidence accumulation, subjective ratings, computational neuropsychology, R

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Introduction

Traditional models can be misleading

Consider a dataset containing reaction times (RTs) from 20 participants tested under two experimental conditions (25 trials each). A traditional analysis approach would involve fitting a linear mixed model testing the effect of condition, with participants entered as random intercepts:

```
lmer(RT ~ Condition + (1 | Participant),
     data = cogmod::badlm)
```

Because this analysis yields no significant difference ($b_{condition} = -0.001$, $t(996) = -0.08$, $p = .94$; Figure 1 B), one could misleadingly interpret and report it as an absence of any difference in the RTs of the two conditions. Yet the two distributions are actually completely different (Figure 1 A): condition A produces a lot of very fast and very slow responses, whereas condition B yields much more consistent RTs. But because the mean of their distribution is similar, the analysis does not capture any other differences. The problem is not the design, the data preprocessing, the random-effect structure or the post-processing, but the model itself: linear regressions estimate means and nothing else. Simply changing the model to use another distribution, such as ex-Gaussian, drastically changes the conclusions that can be drawn from the same data, revealing significant differences in the bulk of the distribution, its spread and its tail (Figure 1 C). Despite the massive benefits we can get from using more appropriate models, they are rarely used in practice, which is what the `cogmod` package aims to address.

From means to mechanisms

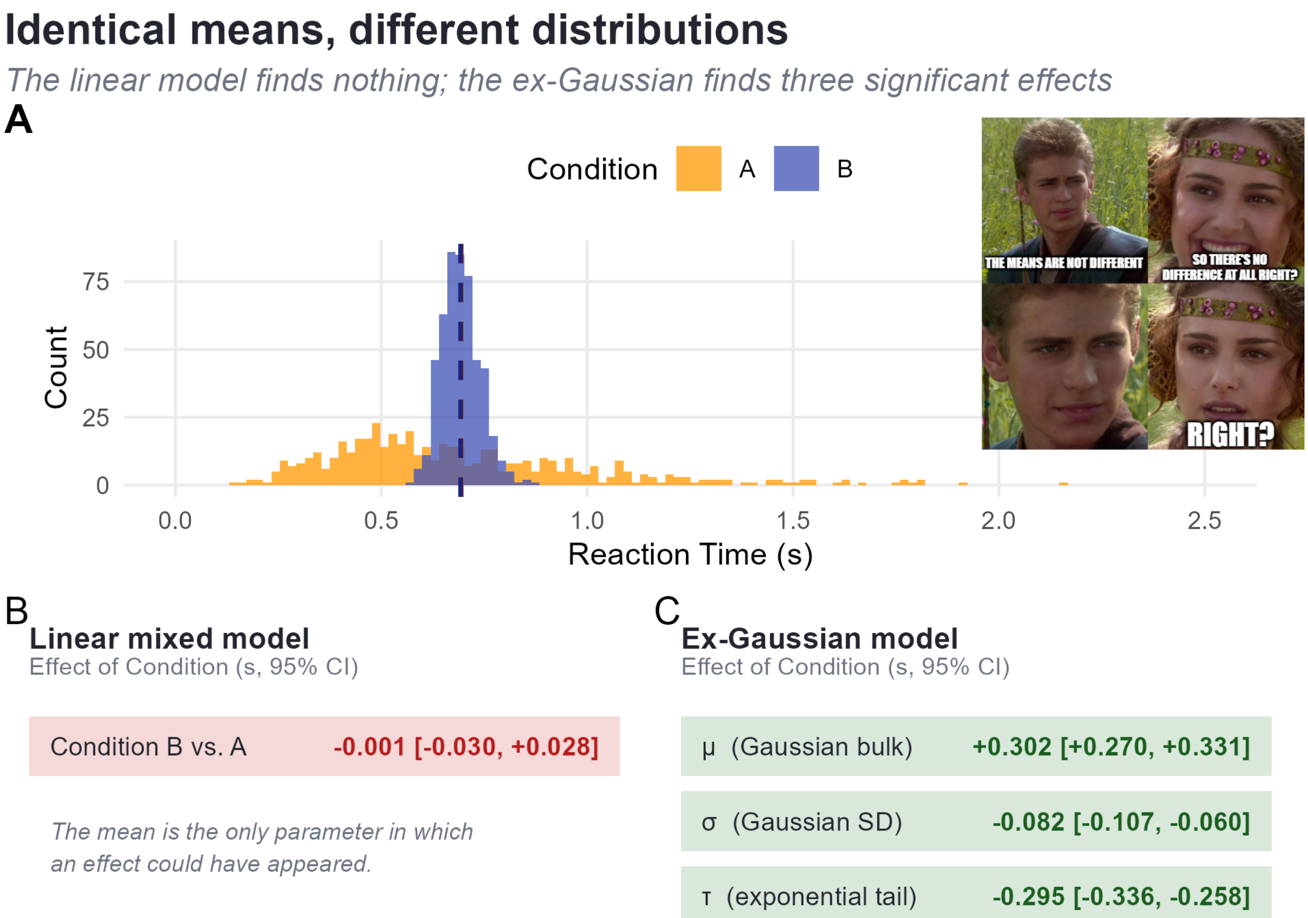
Cognitive tasks yield complex data, resulting from a sequence of processes that unfold over time. Some responses come almost immediately, some follow a lapse of attention, some reflect an unusually cautious pass over the stimulus, and some are emitted before the evidence has really been weighed. The shape of the distribution, its mode, its spread and its tail, is where the underlying processes leave their trace. Most published analyses reduce all of it to a handful of condition means and test for their difference. That reduction is itself a modelling decision, and a costly one: most of what makes cognitive data distinctive is discarded in the process.

The traditional summary-statistics approach involving *t*-tests, ANOVAs, linear regressions, treats observations as Normally distributed around a predicted mean (Figure 2). Two alternatives can be distinguished from it. The **distributional** approach chooses a likelihood that matches the shape of the outcome, such as an ex-Gaussian or shifted LogNormal for reaction times (Balota & Yap, 2011; RTs, Matzke & Wagenmakers, 2009), or an ordered Beta for bounded ratings (Kubinec, 2023), so that skew, boundedness and dispersion are described rather than ignored. The **computational** approach goes further and derives the likelihood from an account of the process that generated the data. Evidence accumulation models (EAMs, or sequential sampling models) treat response times as the first-passage times of a noisy accumulation towards a decision boundary, and their parameters map onto theoretically distinct components of the decision: the quality of the evidence, how much of it is required before committing, and the time spent encoding the stimulus and executing the response (Brown & Heathcote, 2008; Ratcliff & McKoon, 2008).

The case for moving past the first approach is well supported. Distributional analyses recover effects that mean comparisons miss and disambiguate effects that mean comparisons conflate (Balota & Yap, 2011; De Boeck & Jeon, 2019). Process models turn an experimental effect into a statement about *which component of processing changed*, which is what most cognitive-theoretical claims are about (Evans & Wagenmakers, 2020; Ratcliff et al., 2016). Model-based parameters are also more reliable than the difference scores traditionally computed from the same data, making

Figure 1

Simulated reaction times from two experimental conditions, available in the `cogmod` *package.* ***(A)*** *Observed distributions; dashed lines mark the condition means.* ***(B, C)*** *The effect of condition as each model reports it, with 95% intervals; green marks an interval excluding zero, red one that does not. The linear mixed model has only a mean to report, and finds no effect in it. The ex-Gaussian splits the same data into a Gaussian bulk (μ), its spread (σ) and an exponential tail (τ), and finds an effect on all three; because its mean is $\mu + \tau$, the opposing effects on bulk and tail cancel into exactly the null of panel B.*

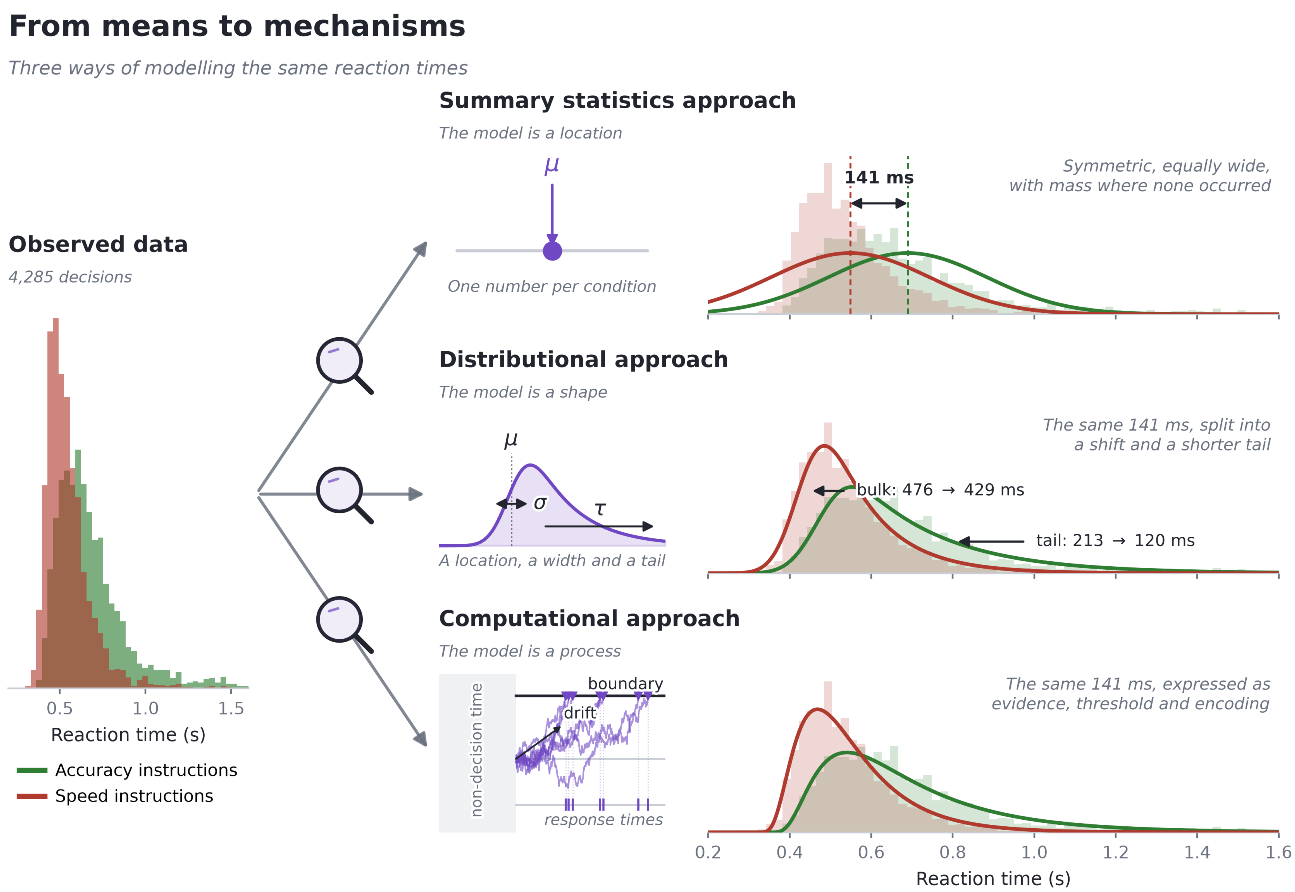


them one of the more promising responses to the "reliability paradox" (Haines et al., 2020; Hedge et al., 2018).

The gap between what a task produces and what is taken from it is widest in neuropsychological assessment, which rests almost entirely on cognitive tasks. The indices extracted from them are total scores, completion times and differences between condition means: quantities inherited for historical rather than theoretical reasons, with no principled mapping onto the constructs they are said to measure (Ratcliff & McKoon, 2022). Older adults, for instance, are reliably slower, but a meta-analysis of diffusion model analyses locates the general age effect on boundary separation and non-decision time, with drift rate differences that are task-dependent

Figure 2

The three approaches applied to the same trials: correct responses from three participants in the lexical decision data of Wagenmakers et al. (2008), used again in Example 1. Observed reaction times are on the left; each row then shows what one model is made of and the distribution that follows from it, drawn back over the same data. The 141 ms difference between instruction conditions is first a single number, then a shift plus a shorter tail, then a statement about how evidence was accumulated. The ex-Gaussian and shifted Wald curves both use posterior means from models fitted to these trials in the package's RT-only Models *vignette.*



and occasionally reversed (Theisen et al., 2021). A latency difference reported as "cognitive slowing" could be more accurately framed as a statement about caution or motor execution. Computational neuropsychology (Steinke & Kopp, 2020) proposes to keep the tasks and replace those scores with parameters that refer to identifiable components of processing, so that a deficit is characterized rather than merely detected.

The same holds for brain-behaviour correlations, a field burdened by weak associations and low reliability. Brain-wide association studies based on basic behavioural indices (such as mean RTs) need thousands of participants before their effects replicate (Marek et al., 2022), and the test-retest reliability of common task-fMRI measures is typically low (Elliott et al., 2020). In contrast, parameters derived from cognitive models are more reliable

(Haines et al., 2020; Rouder & Mehrvarz, 2024) and closer to what a neural account is about. The threshold adjustment made under time pressure is a case in point: it is a model parameter rather than an observable, and seems to accurately track striatal and pre-supplementary motor activation, and cortico-striatal connection strength, across individuals (Forstmann et al., 2008, 2016; Wiecki et al., 2015). Whether the target is a clinical score or a brain-behaviour correlation, the root cause is the same: we only get what we fit, and fitting means with linear models is a self-imposed limitation.

The obstacle is practical

Uptake of alternative approaches remains unfortunately modest, and the obstacle is mostly practical. The software landscape for fitting alternative models is fragmented. Diffusion models are fitted in HDDM or HSSM in Python (Fengler et al., 2026; Wiecki et al., 2013), in fast-dm (Voss & Voss, 2007), in DMAT (Vandekerckhove & Tuerlinckx, 2008), or in PyDDM (Shinn et al., 2020); racing accumulator models in DMC and EMC2 in R (Heathcote et al., 2019; Stevenson et al., 2026), or with `SequentialSamplingModels.jl` in Julia (Fisher & Makowski, 2024); bounded and ordinal responses in yet another set, such as `ordbetareg` (Kubinec, 2025). Each toolbox has its own syntax for specifying which parameters depend on which conditions, its own parameterizations, defaults, and its own output objects.

As a consequence, comparing families is difficult and therefore rare. Deciding between an ex-Gaussian and a shifted LogNormal, or asking whether a Weibull distribution would do as well as either, means re-fitting in different packages under different conventions, which slows exploratory work and leaves the choice of likelihood to be inherited from disciplinary convention rather than tested against the data at hand. Second, these toolboxes often sit apart from the regression workflow psychologists already use, such as `brms` in R (Bürkner, 2017), which means missing on the expressive power of that workflow. Every design has to be restated in a toolbox-specific syntax, and features like complex random effects specifications, smooth terms, monotonic effects for ordinal predictors, or post-processing tools, such as marginal effects and means, are effectively out of reach (Bürkner, 2018; Wood, 2017).

`cogmod` aims to address these limitations by facilitating the application of these models inside a framework researchers already use. Every model is defined as a `brms` custom family (Bürkner, 2017) specified with ordinary `brms` formula syntax. Everything a `brms` user already knows then applies unchanged to a Drift Diffusion Model or an ordered-Beta rating model: distributional regression on every parameter, multilevel structure, priors, convergence diagnostics, posterior predictive checks, leave-one-out cross-validation (Vehtari et al., 2017). The `easystats` ecosystem (Lüdecke et al., 2019, 2021; Makowski et al., 2019) has been adapted to accommodate these families, providing first-rate post-processing tools.

We describe below the design of the package and the models it provides, then give three worked examples, one per family of models, to illustrate the workflow and applications.

Design and Implementation

Models as `brms` custom families

`brms` translates an R formula into Stan code. Its *custom family* mechanism allows users to supply their own likelihood: a `customfamily` object declaring the distributional parameters and their link functions, plus a `stanvars` object injecting the Stan functions that evaluate the log-density. `cogmod` packages both for each model, and additionally provides the post-processing methods (`log_lik_*`, `posterior_predict_*` and `posterior_epred_*`) that `brms` requires in order to compute information criteria and generate predictions used in post-processing.

In practice, using a `cogmod` model requires two arguments beyond a standard `brm()` call: `family` and `stanvars`. Two further helpers, `cogmod_priors()` and `cogmod_inits()`, are strictly speaking optional but should be treated as part of the call: they provide adapted chain-initialization values and weakly informative priors on the sensitive parameters to limit convergence issues and other sampling pathologies.

```r
library(cogmod)
library(brms)

# Specify the formula using brms' bf()
f <- bf(
  RT ~ Condition +
    (Condition | Participant),
  sigma ~ Condition +
    (Condition | Participant),
  ndt ~ Condition +
    (Condition | Participant),
  family = cogmod_lognormal()
)

# Fit the model
m <- brm(
  f,
  data = df,
  prior = cogmod_priors(f, df),
  init = cogmod_inits(f, df),
  stanvars = cogmod_stanvars(f),
  backend = "cmdstanr"
)
```

Everything else is `brms`. The formula syntax is the usual one, so any parameter can be predicted by covariates, given random effects, splines, autocorrelation structures or monotonic effects. The fitted object is a plain `brmsfit`, so `summary()`, `pp_check()`, `loo()`, `emmeans`, `bayesplot` and the `easystats` functions work out of the box. Contrary to many alternative toolboxes, here each distributional parameter is a regression target with its own formula. Testing the effect of a manipulation on various parameters becomes thus trivial, as is giving each participant their own parameters through random effects. This allows the location-scale modelling advocated by Williams et al. (2021), in which the dispersion of a participant's RTs is a person-specific quantity with its own predictors, to be the default rather than a special case.

Relation to existing software

Figure 3 shows the families available at the time of writing. As adding a new model is a self-contained piece of work, the set is expected to grow to track cutting-edge advances in computational psychology. The package's documentation website contains the list of available models, with each family's distributional parameters, link functions and defaults, as well as hands-on tutorials.

`cogmod` is not a replacement for the specialized evidence accumulation toolboxes, and the division of labour is worth stating. EMC2 (Stevenson et al., 2026) and HSSM (Fengler et al., 2026) provide sampling algorithms tailored to the geometry of accumulator models, hierarchical structures designed for them and, in HSSM's case, likelihood approximation networks that make otherwise intractable variants estimable. For a study whose scientific question *is* the architecture of the decision process, those are the right tools, and they will be faster and more capable at the far end of model complexity.

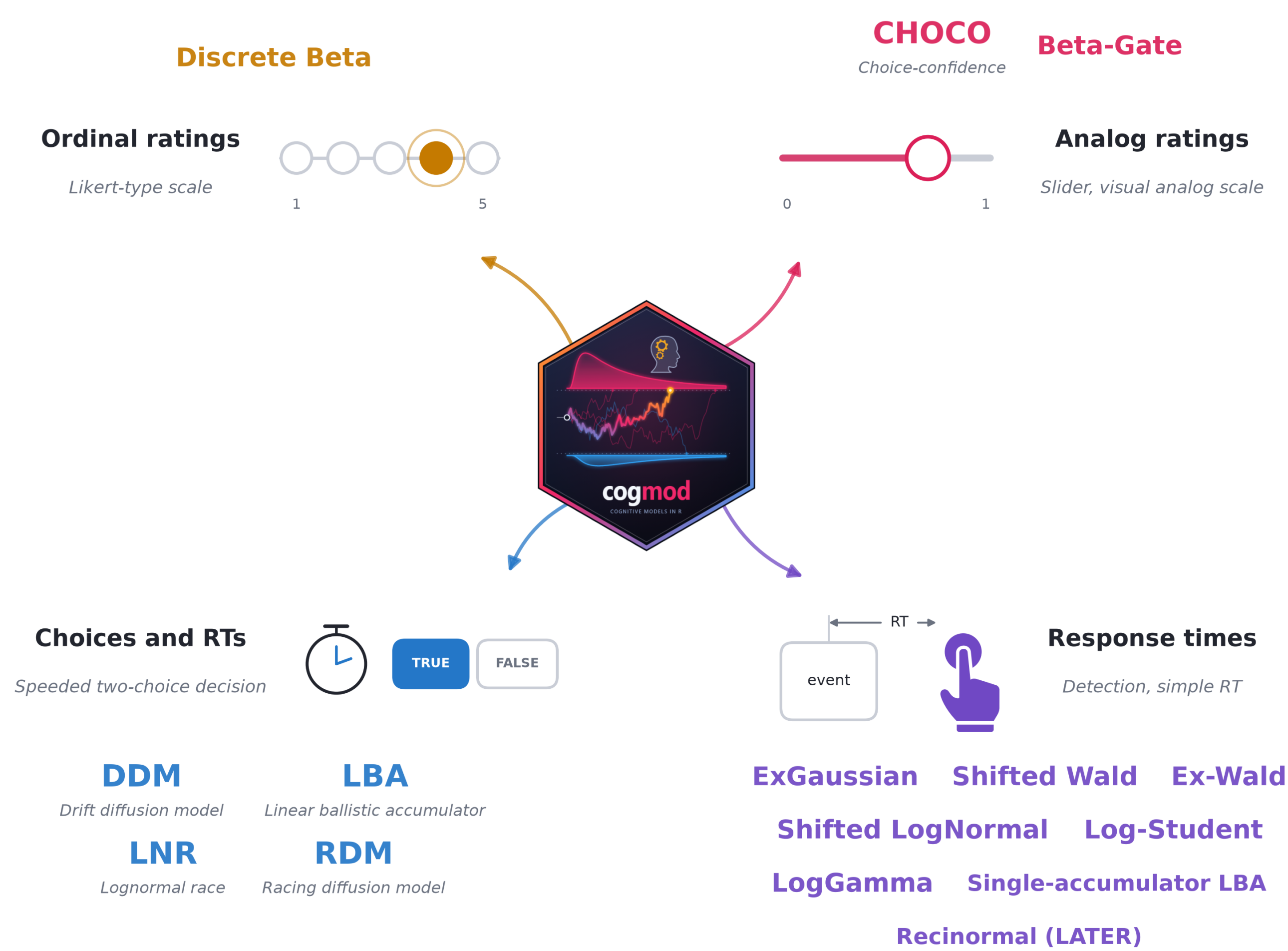
`cogmod` targets a different (and arguably more common) situation: a researcher with an experimental dataset, a regression-shaped question and a need for a likelihood that respects the data. For that user the relevant comparison is between a linear model and something better, not between one accumulator toolbox and another. Moving from `brm(RT ~ Condition)` to a shifted LogNormal or a Wald model means changing two arguments rather than changing software, and at every step the analysis remains a regression, so most of the analyst's expertise and skills continue to apply.

Example 1: Many Decision Models, One Workflow

To illustrate the typical workflow of fitting evidence accumulation models, we will fit them to

Figure 3

Response formats covered by `cogmod` and the main families available for each at the time of writing; the package website carries the current list. Six further descriptive RT families are also available (`cogmod_weibull()`, `cogmod_logweibull()`, `cogmod_invweibull()`, `cogmod_gamma()`, `cogmod_invgamma()` and `cogmod_bisa()`), and the package's RT-only Models *vignette compares them all on the same data. Every family comes with a `_stanvars()` function supplying the corresponding Stan code, and with `d*()` and `r*()` functions for density evaluation and simulation.*



the lexical decision task data of Wagenmakers et al. (2008) (Experiment 1), distributed in the `rtdists` package as `speed_acc`: participants classified letter strings as words or non-words under instructions emphasising either **speed** or **accuracy**. Responses slower than 2 s were excluded. This example uses three participants (n = 4,620 trials, 7.3% of errors), pooled without random effects for clarity and simplicity.

In order to fit a basic Drift Diffusion Model (DDM) in `cogmod`, one needs to specify 1) a formula for each of the parameters, 2) the model family (`cogmod_ddm()`) and attach the custom family Stan code with `cogmod_stanvars()`. As it is common in the field, the "decision" being modelled is **correct vs. error**, which is added to the formula with a `dec()` term on the response side of the formula. Everything else is the ordinary `brms` syntax:

```r
f <- bf(
  RT | dec(Error) ~ Condition,
  boundary ~ Condition,
  bias ~ 1,
  ndt ~ 1,
  sigmadrift = 0,
  sigmabias = 0,
  sigmandt = 0,
  family = cogmod_ddm()
)

m_ddm <- brm(f, data = df,
             prior = cogmod_priors(f, df),
             init = cogmod_inits(f, df),
             stanvars = cogmod_stanvars(f)) |>
  add_criterion("loo")
```

The first unnamed parameter, referred to as `mu` in the Stan code as per `brms` convention, actually corresponds to the drift rate. The other parameters correspond to boundary separation (`boundary`), starting point bias (`bias`), and non-decision time (`ndt`), respectively. Estimating non-decision time is a common source of difficulty in fitting diffusion models: a shifted distribution assigns zero density to any response faster than `ndt`, which puts a hard wall in the likelihood at the fastest observed response. In `cogmod`, every family that carries an `ndt` therefore mixes in a small contaminant component, of weight `poutlier`, that keeps early responses at positive density. The wall becomes a finite cost, and `ndt` is estimated directly in seconds rather than relative to a sample statistic.

The three `sigma*` arguments fix the between-trial variabilities of the full seven-parameter diffusion model (Ratcliff & McKoon, 2008) at zero, giving the "simple" four-parameter DDM, which is a common simplification, since the variability parameters are expensive to sample and notoriously hard to recover (Boehm et al., 2018). The effect of `Condition` is also estimated for the boundary separation, which is the textbook hypothesis for a speed-accuracy manipulation, and the results support it: boundary separation drops sharply under speed instructions (95% CI on the linear predictor $[-0.65, -0.55]$).

Whether that is the right model, however, is a separate question, and answering it means fitting the alternatives, which only involves a few code changes: a different `family`, its `stanvars`, and whichever distributional parameters that family owns. The **DDM-5** is the model above with between-trial variability in drift rate freed (`sigmadrift ~ 1`). The **LogNormal race** (LNR, Rouder et al., 2015), the **linear ballistic accumulator** (LBA, Brown & Heathcote, 2008) and the **racing diffusion model** (RDM, Tillman et al., 2020) replace the single diffusion with two accumulators racing towards a threshold, and differ in how each accumulator gets there: by drawing its finishing time from a LogNormal, by accumulating linearly with between-trial variability in its rate, or by diffusing with within-trial noise alone. Because each fit is a `brmsfit` carrying a `loo` criterion, comparing the five is one call, and Figure 4 B puts the result of that call against what each architecture cost to sample (the ranking itself is three participants' worth of evidence; it is an illustration of the workflow, not a result):

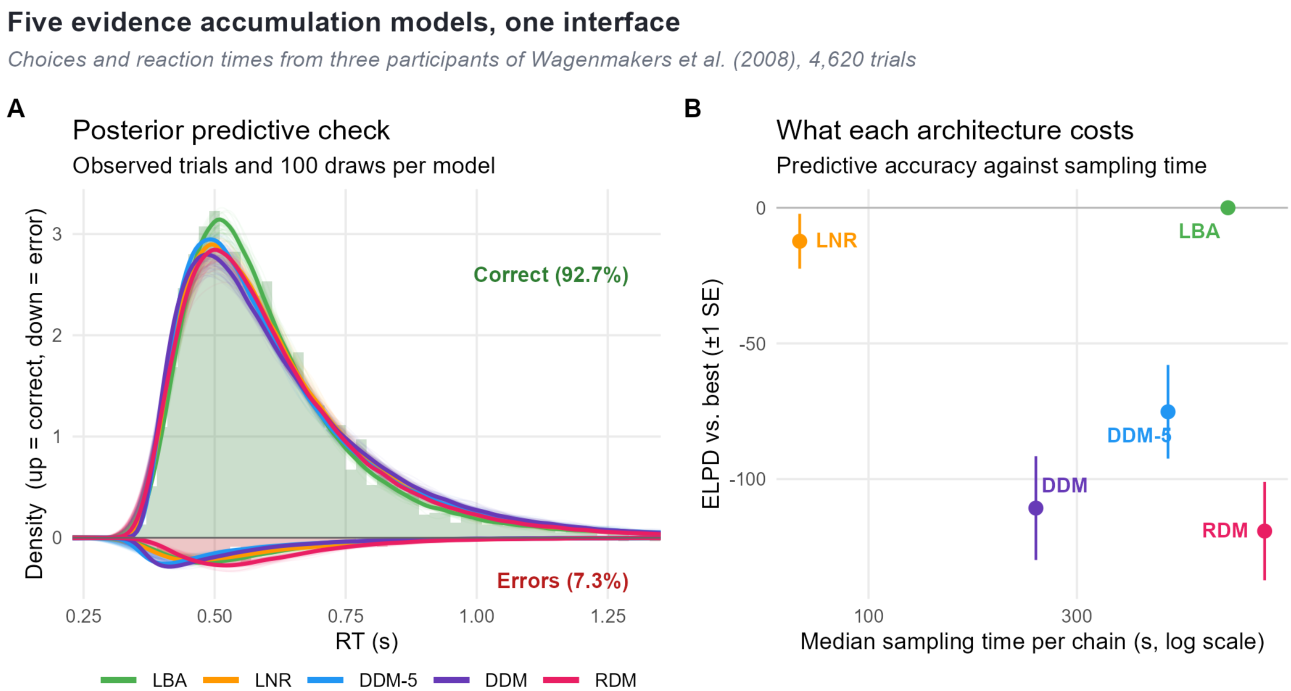
```r
loo_compare(m_ddm, m_ddm5, m_lnr,
            m_lba, m_rdm)
```

Example 2: Parameter Variability

Beside decision-making models, `cogmod` also ships a large collection of RT-only models, spanning the descriptive-to-generative continuum: the ex-Gaussian and the shifted LogNormal, well established in the RT literature, the shifted Wald, the single-accumulator LBA and its recinormal (LATER model) special case whose parameters refer to an accumulation process, and a longer tail of families - log-Student, log-Gamma, Birnbaum-Saunders, and several Weibull and Gamma variants. While they can be easily compared with the same workflow as above, the question this example asks instead concerns interindividual variability.

Figure 4

Five evidence accumulation models fitted to the choices and reaction times of three participants. ***(A)*** *Posterior predictive check: observed trials as histograms, 100 predictive draws per model as faint lines and their pooled density in bold, with correct responses drawn upwards and errors downwards. Both halves are scaled by the* proportion *of that response, so the area under each curve is its predicted frequency and the check is on the joint distribution of choices and RTs. Draws are taken with `with_outliers()`, so the check is on the same mixture the likelihood scores. Predicted RTs above 3 s (0.05% of draws, produced by near-zero drift rates in the races) are excluded.* ***(B)*** *Predictive accuracy against sampling cost. Both panels come out of the ordinary `brmsfit` methods, and neither needed the models to be refitted.*



Consider a simple detection task typical of attentional paradigms: a stimulus appears, the participant responds as fast as they can, with 80 times per three experimental conditions A, B, and C. RT data of thirty people are modelled with a shifted LogNormal mixed model, with every parameter free to vary by condition and by participant (random slopes):

```
f <- bf(
  RT ~ Condition +
    (Condition | Participant),
  sigma ~ Condition +
    (Condition | Participant),
  ndt ~ Condition +
    (Condition | Participant),
  family = cogmod_lognormal()
)
```

Read as an experiment, the result seems clear and straightforward. Condition B significantly slows responses relative to A on the location parameter (B-A: $b_\mu = 0.10$, 95% CI $[0.04, 0.17]$), but condition C is not different (C-A: $b_\mu = -0.02$, 95% CI $[-0.15, 0.10]$). Neither condition impacts the dispersion or the non-decision time. The temptation is to treat the one significant effect as the finding, i.e., the quantity the task “measures” and the natural candidate for a score to correlate with a questionnaire, a phenotype or a scan. It is in fact the *worst* of the six for that purpose, and condition C, the null result, is among the best.

The reason is that a fixed effect and an individual difference are answers to different questions, and a design powered for the first says nothing about the second (Haines et al., 2020; Hedge et al., 2018; Makowski et al., 2023; Rouder & Mehrvarz, 2024). The distinction is decisive wherever the point of a task is to produce a number that characterises a *person* rather than a population - which is to say throughout neuropsychology and differential research, where tasks are routinely adopted on the strength of a group effect alone. The random-effect SDs already hint at it: 0.01 for the condition B slope against 0.28 for the condition C slope, more than twenty times larger. Condition B slows *everyone* by about the same amount, which is exactly what makes its group effect so precise. Condition C slows some participants dramatically and speeds others up just as much, which averages to nothing.

Participant-level estimates can be extracted with `modelbased::estimate_grouplevel()`, and their usability as individual scores can be summarised by the **Variance-Over-Uncertainty Ratio** (D-vour), implemented in `performance::performance_dvour()` (Lüdecke et al., 2021):

$$D_{\text{vour}} = \frac{\sigma^2_{\text{between}}}{\sigma^2_{\text{between}} + \mu^2_{\text{within}}}$$

where $\sigma^2_{\text{between}}$ is the observed variability of the participant-level estimates and μ^2_{within} their mean squared uncertainty. It is bounded in $[0, 1]$, and values near 1 mean participants differ far more than we are uncertain about where each of them stands. It is close to an intraclass correlation, but its denominator uses the *uncertainty of the point-estimates* rather than the raw within-participant variance, so that, unlike an ICC, collecting more trials per participant increases it; and relates to the signal-to-noise ratio advocated by Rouder and Mehrvarz (2024).

Figure 5 shows the result for every parameter of the model. The intercept (the location parameter of Condition A) and condition C both have a broad marginal band and points that fan out well past their intervals, which means people are well separated on these parameters. In contrast, the effect of Condition B vs. A (the only significant fixed effect), has points around zero inside intervals that dwarf their spread.

While this simulated data was created specifically with no inter-individual variability in the `sigma` and `ndt` parameters, real data often look different: RT variability can often discriminate between individuals better than the mean does (Williams et al., 2021), and non-decision time is a natural candidate for a stable person-specific quantity. But in order to be able to assess that, one needs to fit appropriate models. The recommendation is simple: to model and quantify variability for every parameter, and report it alongside the "main" effects.

Example 3: Illustration in a Clinical Population

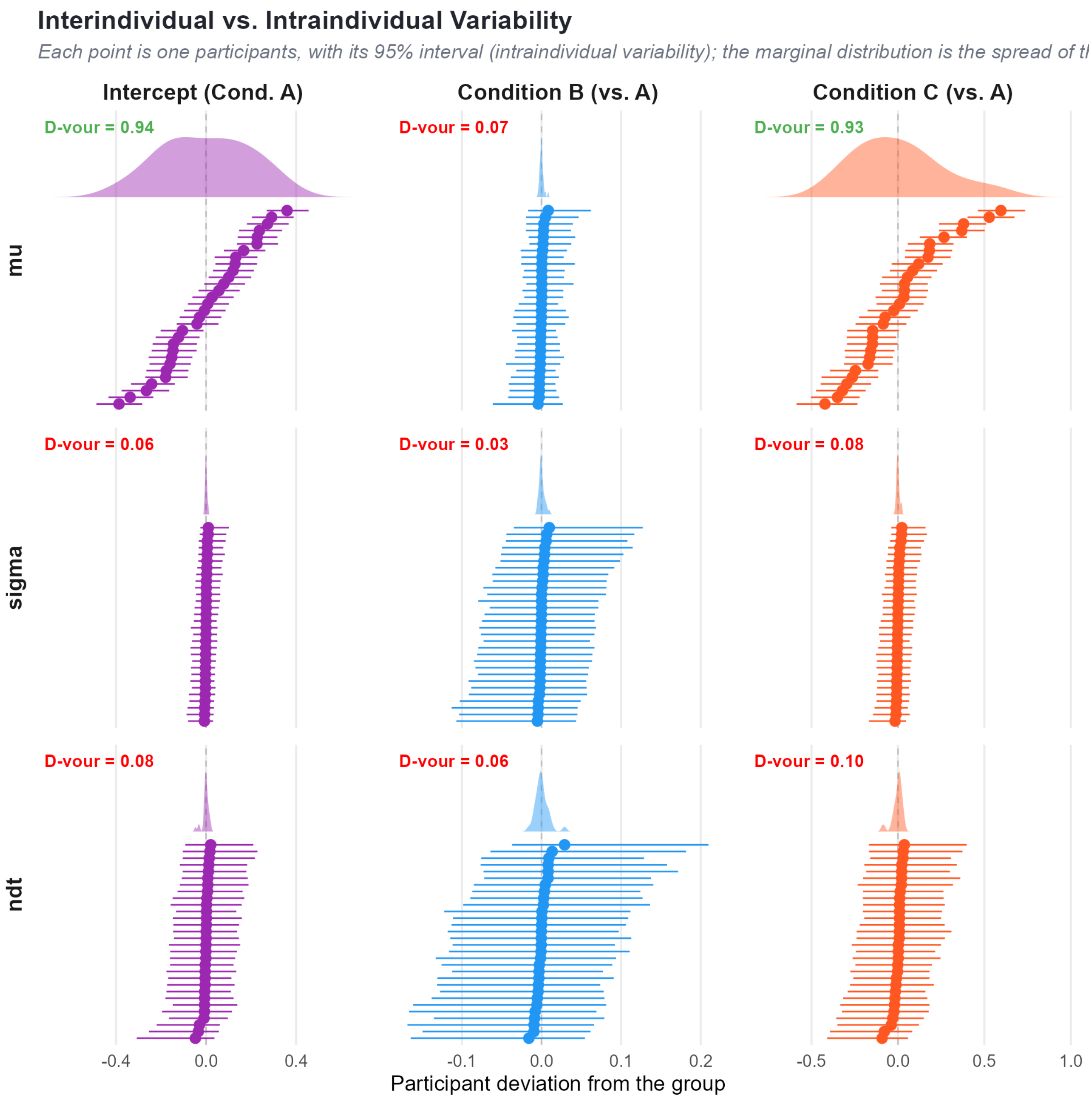
The data so far have been a laboratory experiment and a simulation. The case for computational neuropsychology, though, is illustrated with clinical data, where trials are few, samples are heterogeneous and where more appropriate models can significantly improve insights gained from the data.

We showcase this on the UCLA Consortium for Neuropsychiatric Phenomics dataset (Poldrack et al., 2016), openly available on OpenNeuro as `ds000030`, and take the task-switching data from 42 participants diagnosed with ADHD and 125 healthy controls. Participants classified a coloured shape by either its colour or its shape, with the relevant dimension cued on each trial and switching on a quarter of them. Keeping correct responses between 200 ms and 2 s leaves 14,543 trials, at most 96 per participant.

The standard index for this task is the *switch cost*, the difference between a participant's mean RT on

Figure 5

Individual-level estimates for every parameter of the mixed shifted LogNormal. Each panel is one parameter: points are the 30 participants' deviations from the group, ordered by estimate, with 95% intervals; the shaded band above is the marginal distribution of those point estimates. A parameter is usable as an individual score when the points spread further than the intervals are wide - which is what D-vour, printed in each panel, measures. Green marks values above 0.5. `estimate_grouplevel()` *labels the* `mu` *block "conditional", since it is the response formula; it is written as* `mu` *here to match the family's parameter names.*



switch and non-switch trials, read as a measure of executive control. A t-test shows no significant difference between the groups: Controls average 111 ms and the ADHD group 113 ms, $t(82) = -0.13$, $p = .90$. A linear mixed model agrees: a large switch cost overall ($b = 0.11$ s, $t = 13.3$, $p < .001$, so the manipulation worked), no group difference in it ($p = .89$), and no reliable group difference in overall speed ($p = .12$). Taken at face value, this task detects no deficit in this sample.

We re-analyzed this dataset using a shifted LogNormal model in which diagnosis and switching predict the location, the dispersion and the nondecision time, with random effects by participant:

```
f <- bf(
  RT ~ Diagnosis * Switching +
    (Switching | Participant),
  sigma ~ Diagnosis * Switching +
    (1 | Participant),
  ndt ~ Diagnosis + (1 | Participant),
  family = cogmod_lognormal()
)
```

The model agrees about the switch cost: the interaction is null on the location and on the dispersion. But diagnosis registers on all three parameters at once, and in opposing directions. On the response scale (Figure 6), the ADHD group's location parameter is 96 ms slower (95% CI $[26, 178]$), their non-decision time is 48 ms faster (95% CI $[-75, -16]$), and their log-scale dispersion is lower (95% CI $[-64.9, 4.4]$). The two effects largely cancel, leaving the total mean difference nonsignificant (95% CI $[-17, 124]$), which is what the null the t-test reported.

This is the cancellation of Figure 1, present in real clinical data: two components move in opposite directions, the mean records their sum, and a difference score computed from means is therefore blind to both. No amount of statistical power on the switch cost would have recovered this.

Note however that the decision and non-decision parameters trade off in the posterior ($r = -0.44$), so the split between them is less certain than either interval taken alone; what is robust is that a group difference exists in the shape of the distributions and not in their means. And this is one task in one sample, analysed post hoc: it is a demonstration that the question becomes askable, not a claim about the neuropsychology of ADHD.

Examples 2 and 3 are two halves of one clinical argument. A group difference in a parameter, as here, is a statement about a population; using that parameter as a patient's score requires the check of Example 2, run on the same fit. Both come out of one model, and neither requires the analyst to leave the regression model they started from.

Example 4: Subjective Ratings

Cognitive modelling is usually applied to RTs, but the principles apply to subjective ratings as well, which exhibit particular distribution shapes worth characterizing. Analog (slider) scales are bounded, and responses pile up at the endpoints, because answering "completely agree" is a different kind of act from answering "agree quite a lot". A mean rating cannot tell the two apart.

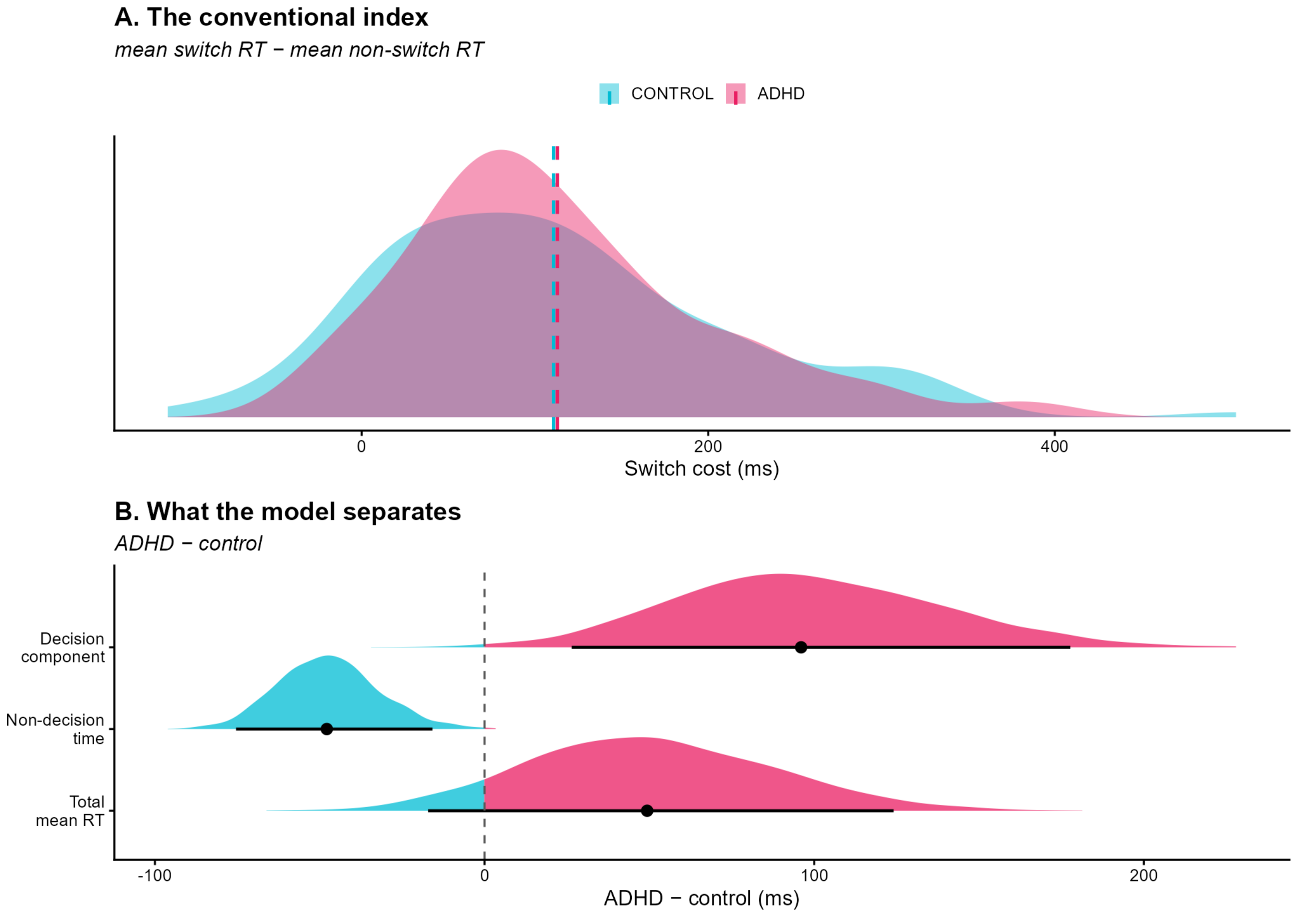
Consider a taste test. Participants rate how much they enjoyed each of two dishes on a slider running from "hated it" to "loved it". One dish is a margherita pizza: few people have strong feelings about it[1], and the ratings pile up in the middle. The other is a coriander salad, which a substantial minority of people experience as tasting of soap. Half the room loves that dish and half hates it, and almost nobody is indifferent.

Similarly to the examples above, a t-test duly reports that they are equally liked ($t(296) = 0$, $p = 1.00$), and a linear regression agrees, putting the difference at 0.00 with a 95% CI of $[-0.05, 0.05]$. `cogmod_betagate()` is `cogmod`'s reparameterization of the ordered Beta model (Kubinec, 2023), in which an observed 0 or 1 arises when an underlying continuous tendency passes a lower or upper "gate".

[1] Data from the author were not included, as they would have biased the margherita mean towards extreme enjoyment.

Figure 6

The same task-switching data under two analyses (42 ADHD, 125 controls). (A) The conventional switch-cost index, computed for each participant as their mean RT on switch trials minus their mean RT on non-switch trials. The distributions of these individual switch costs do not separate the groups. Dashed lines mark the group means. (B) Posterior distributions of the ADHD - control differences under the shifted LogNormal model, on the response scale. The total mean RT difference is close to zero because a slower decision component is accompanied by a faster non-decision component.



Its four parameters are separately interpretable: `mu` is that underlying mean tendency, `phi` the rating's consistency (the spread), `pex` the overall probability of an extreme response, and `bex` the bias in the direction of the extremity. As in every other family, each takes its own formula:

```
f <- bf(score ~ Dish,
        phi ~ Dish,
        pex ~ Dish,
        bex ~ Dish,
        family = cogmod_betagate())

m <- brm(f, data = df,
         stanvars = cogmod_stanvars(f))
```

The four parameters separate what the mean confounded (Figure 7 C). Consistency collapses, from 11.2 to 0.5, so the coriander ratings are not merely more variable but *anti*-clustered ($\phi < 1$): mass at both ends rather than in the middle. Extreme responding (zeros and ones) rises from 3% to 41% of trials. And the two parameters that should not move do not: the underlying mean tendency is unchanged (+0.02, $[-0.03, 0.08]$), and so is the direction of the extremes (-0.03, $[-0.39, 0.31]$), since the coriander extremes split evenly between "hated it" and "loved it" rather than favouring one end.

This example generalises past taste tests. Consensus and polarisation are routinely confounded in exactly this way whenever attitudes, symptom severity or confidence are summarised by an average. Beta-Gate is one of several ways to give a bounded response its point masses, and the zero-one inflated Beta (ZOIB) and the extended-support Beta (Kosmidis & Zeileis, 2024) are others, and each attributes the endpoints to a different mechanism. Discrete rating scales are covered too: `cogmod_betadiscrete()` applies the same logic to Likert-type items, where the response is one of a handful of ordered categories rather than a point on a continuum. A response format that ordinarily calls for its own specialised package is here one more `family` argument, carrying the same formula syntax, the same diagnostics and the same post-processing as any other `brms` models.

Discussion

`cogmod` implements Bayesian cognitive models as `brms` custom families: descriptive RT distributions, evidence accumulation models, and models for bounded and Likert-type ratings. The design commitment is minimal. The package supplies likelihoods and the post-processing methods to make them work within the `brms` ecosystem, and inherits the rest.

The examples thread around one important *fil rouge*: *what you fit is what you get*. Choosing a given model is not only a claim about the shape of the data; it fixes the vocabulary in which results can be stated. While a Gaussian model can report that two conditions' mean differ by 141 ms, an ex-Gaussian can attribute that difference to the bulk or to the tail, but a diffusion model will provide answers in terms of evidence, threshold or encoding.

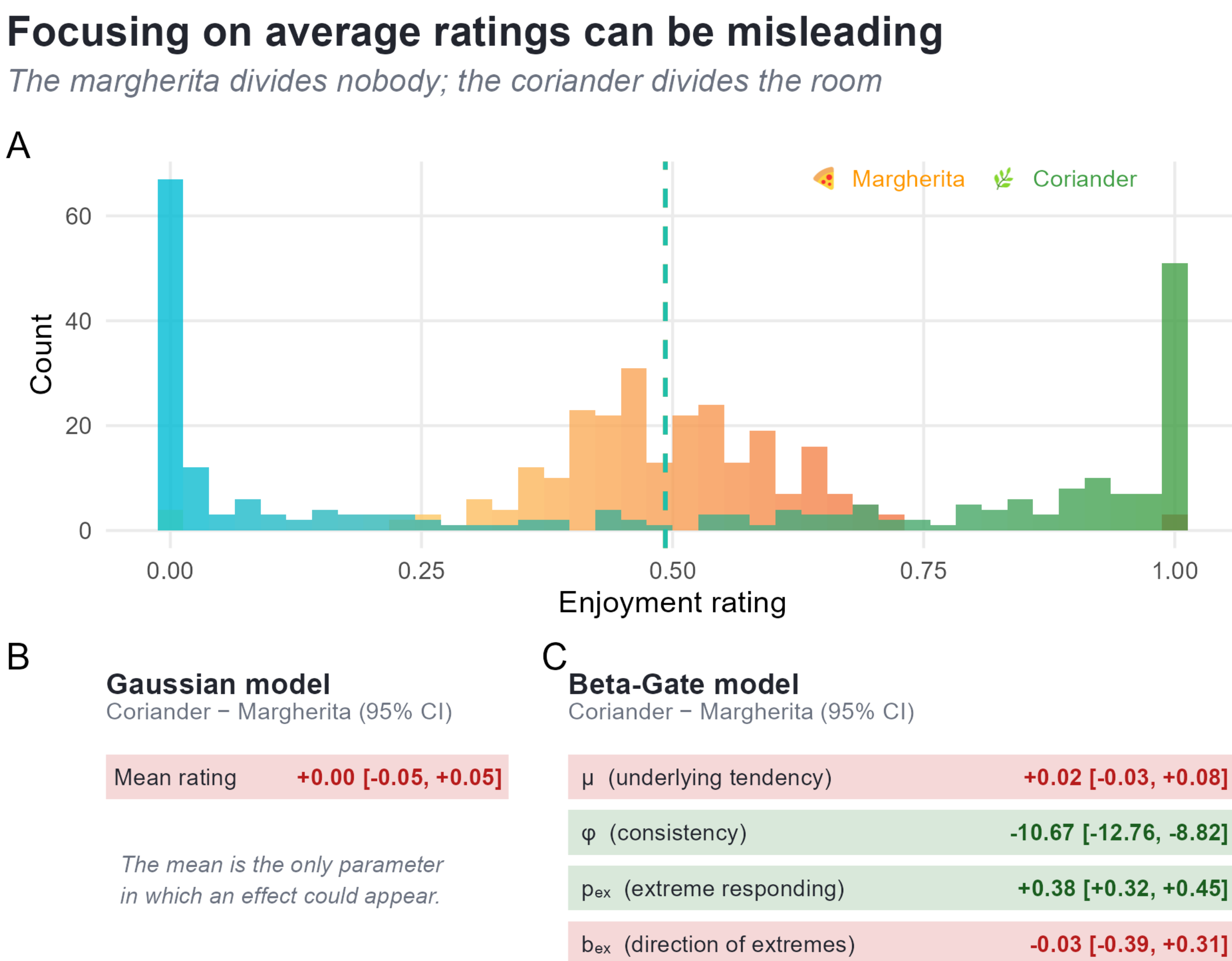
This point is particularly relevant for neuropsychological assessments. A neuropsychological test is a cognitive task, so everything above applies to it directly, with higher stakes attached. The quantity a clinician reports, e.g., a completion time, a total score, an interference effect, is exactly the aggregate that the opening example shows to be several things at once, and a patient's deviation from a normative sample can therefore be difficult to interpret. Model parameters turn "impaired on the Stroop" into a claim about which component is impaired, and two patients with the same score but opposite parameter profiles stop being indistinguishable (Ratcliff & McKoon, 2022; Steinke & Kopp, 2020). Ageing shows what this changes in practice (Theisen et al., 2021), and the clinical reanalysis of Example 3 shows the sharper version of the same point: a conventional index can report no group difference at all while two components underneath it differ substantially.

The reliability check that precedes it supplies the rest of the argument. A parameter is usable as a clinical score only if it separates people by more than it is uncertain about each of them, and this framework makes that checkable for every parameter rather than assumed for whichever one the test happens to report. The same approach helps on the neural side. Correlating brain measures with mean RT or accuracy asks the brain to explain a quantity that mixes mechanisms; correlating them with drift rate, threshold or non-decision time asks it to explain one at a time, which is why parameters have proven the more tractable targets in model-based cognitive neuroscience (Forstmann et al., 2016; Wiecki et al., 2015).

Advanced computational modelling of Human behaviour remained for a long time a relatively niche field of research, difficult to access, adopt, and apply. `cogmod` aims to make these tools accessible to a wider audience, to improve methodological practices with positive benefits for (neuro)psychological science.

Figure 7

The same 500 ratings under two models. ***(A)*** *Observed ratings for the two dishes; the dashed lines mark the condition means, which coincide.* ***(B, C)*** *The effect of dish as each model reports it, with 95% intervals; green marks an interval excluding zero, red one that does not. The Gaussian model has only a mean to report and finds nothing in it. Beta-Gate separates how highly the dish is rated from how* consistently, *and from how often raters go to an endpoint - and it correctly reports no change in the first or in which endpoint they choose.*



Limitations

A few limitations and caveats have to be noted, the first concerning performance. General-purpose HMC on custom Stan code is slower than a sampler built for a specific model class. The evidence accumulation models are particularly slow, and hierarchical versions cost substantially more. These costs can be mitigated in several ways. Stan's within-chain parallelisation and distributed workflows can be exploited on high-performance computing clusters. Alternative Bayesian inference approaches such as variational inference can trade some of the guarantees or generality of full HMC for substantially lower computational cost.

Second, the flexibility of the `brms` interface makes it easy to write down models the data cannot support. Regressing every parameter of an LBA on every condition is one line of code that might appear appealing but is usually a bad idea: evidence accumulation parameters trade off against each other, and between-trial variability parameters are notoriously poorly identified (Boehm et al., 2018; Evans & Wagenmakers, 2020). The package does not, and cannot, prevent this. Prior predictive checks, parameter recovery on simulated data and the standard Bayesian workflow (Gelman et al., 2020) remain the user's responsibility.

The last point is a technical boundary on what `cogmod` can become, and it follows directly from the design. A `brms` custom family is a Stan model, so every family shipped here needs a likelihood that can be written down in closed form and evaluated cheaply enough for HMC. Several of the models a user might reasonably want next do not have one: the leaky competing accumulator (Usher & McClelland, 2001), diffusion processes with collapsing boundaries or time-varying drift (Hawkins et al., 2015), and richer racing architectures are defined by a generative simulation rather than by a density, and where a density does exist it often requires numerical integration, which gradient-based sampling tolerates poorly. The general remedy is to stop requiring the likelihood. Simulation-based inference (Cranmer et al., 2020) trains a neural network on data simulated from the model, either to approximate the likelihood (Fengler et al., 2021, 2026), or to amortize the posterior directly, as in BayesFlow (Radev et al., 2020, 2023). That literature is moving quickly and `cogmod` follows it closely; whether such approximations can be exposed behind the same formula interface, so that a simulation-defined model is specified, sampled and post-processed like any other family, is an open question and the direction in which we hope to extend the package.

Conclusion

The gap between what cognitive data contain and what is extracted from them is large, and it is sustained by friction rather than disagreement. Few researchers would defend a mean difference as an adequate summary of a reaction time distribution, yet many report one, because the alternative often involves learning a new complex set of tools. By implementing cognitive models as families within a regression framework psychologists already use, `cogmod` aims to make the better analysis the easier one; by shipping various descriptive and generative models under one interface, it aims to make the choice between them an empirical question rather than an inherited convention.

Open Practices Statement

`cogmod` is open source and available at https://github.com/DominiqueMakowski/cogmod, with documentation at https://dominiquemakowski.github.io/cogmod/.

Declarations

Funding

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Conflicts of interest/Competing interests

The authors have no relevant financial or non-financial interests to disclose.

Ethics approval

Not applicable. Example 3 re-analyses openly available, de-identified, previously published third-party data (Poldrack et al., 2016); no new data involving human participants were collected for this manuscript.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

The clinical dataset re-analysed in Example 3 is openly available on OpenNeuro as `ds000030` (Poldrack et al., 2016). All other examples use simulated data generated by code included in the package repository.

Code availability

All code, including the `cogmod` package source and the scripts used to generate the examples and figures in this manuscript, is openly available at https://github.com/DominiqueMakowski/cogmod.

References

Balota, D. A., & Yap, M. J. (2011). Moving beyond the mean in studies of mental chronometry: The power of response time distributional analyses. *Current Directions in Psychological Science*, *20*(3), 160–166. https://doi.org/10.1177/0963721411408885

Boehm, U., Annis, J., Frank, M. J., Hawkins, G. E., Heathcote, A., Kellen, D., Krypotos, A.-M., Lerche, V., Logan, G. D., Palmeri, T. J., et al. (2018). Estimating across-trial variability parameters of the diffusion decision model: Expert advice and recommendations. *Journal of Mathematical Psychology*, *87*, 46–75. https://doi.org/10.1016/j.jmp.2018.09.004

Brown, S. D., & Heathcote, A. (2008). The simplest complete model of choice response time: Linear ballistic accumulation. *Cognitive Psychology*, *57*(3), 153–178. https://doi.org/10.1016/j.cogpsych.2007.12.002

Bürkner, P.-C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software*, *80*(1), 1–28. https://doi.org/10.18637/jss.v080.i01

Bürkner, P.-C. (2018). Advanced Bayesian multilevel modeling with the R package brms. *The R Journal*, *10*(1), 395–411. https://doi.org/10.32614/RJ-2018-017

Cranmer, K., Brehmer, J., & Louppe, G. (2020). The frontier of simulation-based inference. *Proceedings of the National Academy of Sciences*, *117*(48), 30055–30062. https://doi.org/10.1073/pnas.1912789117

De Boeck, P., & Jeon, M. (2019). An overview of models for response times and processes in cognitive tests. *Frontiers in Psychology*, *10*, 102. https://doi.org/10.3389/fpsyg.2019.00102

Elliott, M. L., Knodt, A. R., Ireland, D., Morris, M. L., Poulton, R., Ramrakha, S., Sison, M. L., Moffitt, T. E., Caspi, A., & Hariri, A. R. (2020). What is the test-retest reliability of common task-functional MRI measures? New empirical evidence and a meta-analysis. *Psychological Science*, *31*(7), 792–806. https://doi.org/10.1177/0956797620916786

Evans, N. J., & Wagenmakers, E.-J. (2020). Evidence accumulation models: Current limitations and future directions. *The Quantitative Methods for Psychology*, *16*(2), 73–90. https://doi.org/10.20982/tqmp.16.2.p073

Fengler, A., Govindarajan, L. N., Chen, T., & Frank, M. J. (2021). Likelihood approximation networks (LANs) for fast inference of simulation models in cognitive neuroscience. *eLife*, *10*, e65074. https://doi.org/10.7554/eLife.65074

Fengler, A., Xu, P., Bera, K., Paniagua, C., Omar, A., & Frank, M. J. (2026). HSSM: A widely applicable toolbox for hierarchical Bayesian neurocognitive modeling. *bioRxiv*. https://doi.org/10.1101/2026.06.05.730398

Fisher, C. R., & Makowski, D. (2024). SequentialSamplingModels.jl: A julia package for simulating and evaluating sequential sampling models. *Proceedings of the JuliaCon Conferences*. https://doi.org/10.21105/jcon.00186

Forstmann, B. U., Dutilh, G., Brown, S., Neumann, J., von Cramon, D. Y., Ridderinkhof, K. R., & Wagenmakers, E.-J. (2008). Striatum and pre-SMA facilitate decision-making under time pressure. *Proceedings of the National Academy of Sciences*, *105*(45), 17538–17542. https://doi.org/10.1073/pnas.0805903105

Forstmann, B. U., Ratcliff, R., & Wagenmakers, E.-J. (2016). Sequential sampling models in cognitive neuroscience: Advantages, applications, and extensions. *Annual Review of Psychology*, *67*, 641–666. https://doi.org/10.1146/annurev-psych-122414-033645

Gelman, A., Vehtari, A., Simpson, D., Margossian, C. C., Carpenter, B., Yao, Y., Kennedy, L., Gabry, J., Bürkner, P.-C., & Modrák, M. (2020). Bayesian workflow. *arXiv Preprint*, *arXiv:2011.01808*. https://doi.org/10.48550/arXiv.2011.01808

Haines, N., Kvam, P. D., Irving, L. H., Smith, C., Beauchaine, T. P., Pitt, M. A., Ahn, W.-Y., & Turner, B. M. (2020). *Theoretically informed generative models can advance the psychological and brain sciences: Lessons from the reliability paradox*. PsyArXiv. https://doi.org/10.31234/osf.io/xr7y3

Hawkins, G. E., Forstmann, B. U., Wagenmakers, E.-J., Ratcliff, R., & Brown, S. D. (2015). Revisiting the evidence for collapsing boundaries and urgency signals in perceptual decision-making. *Journal of Neuroscience*, *35*(6), 2476–2484. https://doi.org/10.1523/JNEUROSCI.2410-14.2015

Heathcote, A., Lin, Y.-S., Reynolds, A., Strickland, L., Gretton, M., & Matzke, D. (2019). Dynamic models of choice. *Behavior Research Methods*, *51*(2), 961–985. https://doi.org/10.3758/s13428-018-1067-y

Hedge, C., Powell, G., & Sumner, P. (2018). The reliability paradox: Why robust cognitive tasks do not produce reliable individual differences. *Behavior Research Methods*, *50*(3), 1166–1186. https://doi.org/10.3758/s13428-017-0935-1

Kosmidis, I., & Zeileis, A. (2024). Extended-support beta regression for $[0, 1]$ responses. *arXiv Preprint*, *arXiv:2409.07233*. https://doi.org/10.48550/arXiv.2409.07233

Kubinec, R. (2023). Ordered beta regression: A parsimonious, well-fitting model for continuous data with lower and upper bounds. *Political Analysis*, *31*(4), 519–536. https://doi.org/10.1017/pan.2022.20

Kubinec, R. (2025). *Ordbetareg: Ordered beta regression models with 'brms'*. https://doi.org/10.32614/CRAN.package.ordbetareg

Lüdecke, D., Ben-Shachar, M. S., Patil, I., Waggoner, P., & Makowski, D. (2021). performance: An R package for assessment, comparison and testing of statistical models. *Journal of Open Source Software*, *6*(60), 3139. https://doi.org/10.21105/joss.03139

Lüdecke, D., Waggoner, P., & Makowski, D. (2019). insight: A unified interface to access information from model objects in R. *Journal of Open Source Software*, *4*(38), 1412. https://doi.org/10.21105/joss.01412

Makowski, D., Ben-Shachar, M. S., & Lüdecke, D. (2019). bayestestR: Describing effects and their uncertainty, existence and significance within the Bayesian framework. *Journal of Open Source Software*, *4*(40), 1541. https://doi.org/10.21105/joss.01541

Makowski, D., Chen, S.-H. A., Larsen, R. R., Boyle, G. J., & Lilienfeld, S. O. (2023). Clinical neuropsychology in the era of neuroimaging. In *The SAGE Handbook of Clinical Neuropsychology: Clinical Neuropsychological Disorders* (p. 41). SAGE.

Marek, S., Tervo-Clemmens, B., Calabro, F. J., Montez, D. F., Kay, B. P., Hatoum, A. S., Donohue, M. R., Foran, W., Miller, R. L., Hendrickson, T. J., et al. (2022). Reproducible brain-wide association studies require thousands of individuals. *Nature*, *603*(7902), 654–660. https://doi.org/10.1038/s41586-022-04492-9

Matzke, D., & Wagenmakers, E.-J. (2009). Psychological interpretation of the ex-Gaussian and shifted Wald parameters: A diffusion model analysis. *Psychonomic Bulletin & Review*, *16*(5), 798–817. https://doi.org/10.3758/PBR.

16.5.798

Poldrack, R. A., Congdon, E., Triplett, W., Gorgolewski, K. J., Karlsgodt, K. H., Mumford, J. A., Sabb, F. W., Freimer, N. B., London, E. D., Cannon, T. D., & Bilder, R. M. (2016). A phenome-wide examination of neural and cognitive function. *Scientific Data*, *3*, 160110. https://doi.org/10.1038/sdata.2016.110

Radev, S. T., Mertens, U. K., Voss, A., Ardizzone, L., & Köthe, U. (2020). BayesFlow: Learning complex stochastic models with invertible neural networks. *IEEE Transactions on Neural Networks and Learning Systems*, *33*(4), 1452–1466. https://doi.org/10.1109/TNNLS.2020.3042395

Radev, S. T., Schmitt, M., Schumacher, L., Elsemüller, L., Pratz, V., Schälte, Y., Köthe, U., & Bürkner, P.-C. (2023). BayesFlow: Amortized Bayesian workflows with neural networks. *Journal of Open Source Software*, *8*(89), 5702. https://doi.org/10.21105/joss.05702

Ratcliff, R., & McKoon, G. (2008). The diffusion decision model: Theory and data for two-choice decision tasks. *Neural Computation*, *20*(4), 873–922. https://doi.org/10.1162/neco.2008.12-06-420

Ratcliff, R., & McKoon, G. (2022). Can neuropsychological testing be improved with model-based approaches? *Trends in Cognitive Sciences*, *26*(11), 899–901. https://doi.org/10.1016/j.tics.2022.08.015

Ratcliff, R., Smith, P. L., Brown, S. D., & McKoon, G. (2016). Diffusion decision model: Current issues and history. *Trends in Cognitive Sciences*, *20*(4), 260–281. https://doi.org/10.1016/j.tics.2016.01.007

Rouder, J. N., & Mehrvarz, M. (2024). A psychometrics of individual differences in experimental tasks. *Current Directions in Psychological Science*. https://doi.org/10.1177/09637214231220923

Rouder, J. N., Province, J. M., Morey, R. D., Gomez, P., & Heathcote, A. (2015). The lognormal race: A cognitive-process model of choice and latency with desirable psychometric properties. *Psychometrika*, *80*(2), 491–513. https://doi.org/10.1007/s11336-013-9396-3

Shinn, M., Lam, N. H., & Murray, J. D. (2020). A flexible framework for simulating and fitting generalized drift-diffusion models. *eLife*, *9*, e56938. https://doi.org/10.7554/eLife.56938

Steinke, A., & Kopp, B. (2020). Toward a computational neuropsychology of cognitive flexibility. *Brain Sciences*, *10*(12), 1000. https://doi.org/10.3390/brainsci10121000

Stevenson, N., Donzallaz, M. C., Innes, R. J., Forstmann, B. U., Matzke, D., & Heathcote, A. (2026). Bayesian hierarchical cognitive modeling with the EMC2 package. *Behavior Research Methods*. https://doi.org/10.3758/s13428-025-02869-y

Theisen, M., Lerche, V., von Krause, M., & Voss, A. (2021). Age differences in diffusion model parameters: A meta-analysis. *Psychological Research*, *85*(5), 2012–2021. https://doi.org/10.1007/s00426-020-01371-8

Tillman, G., Van Zandt, T., & Logan, G. D. (2020). Sequential sampling models without random between-trial variability: The racing diffusion model of speeded decision making. *Psychonomic Bulletin & Review*, *27*(5), 911–936. https://doi.org/10.3758/s13423-020-01738-8

Usher, M., & McClelland, J. L. (2001). The time course of perceptual choice: The leaky, competing accumulator model. *Psychological Review*, *108*(3), 550–592. https://doi.org/10.1037/0033-295X.108.3.550

Vandekerckhove, J., & Tuerlinckx, F. (2008). Diffusion model analysis with MATLAB: A DMAT primer. *Behavior Research Methods*, *40*(1), 61–72. https://doi.org/10.3758/BRM.40.1.61

Vehtari, A., Gelman, A., & Gabry, J. (2017). Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. *Statistics and Computing*, *27*(5), 1413–1432. https://doi.org/10.1007/s11222-016-9696-4

Voss, A., & Voss, J. (2007). Fast-dm: A free program for efficient diffusion model analysis.

Behavior Research Methods, *39*(4), 767–775. https://doi.org/10.3758/BF03192967

Wagenmakers, E.-J., Ratcliff, R., Gomez, P., & McKoon, G. (2008). A diffusion model account of criterion shifts in the lexical decision task. *Journal of Memory and Language*, *58*(1), 140–159. https://doi.org/10.1016/j.jml.2007.04.006

Wiecki, T. V., Poland, J., & Frank, M. J. (2015). Model-based cognitive neuroscience approaches to computational psychiatry: Clustering and classification. *Clinical Psychological Science*, *3*(3), 378–399. https://doi.org/10.1177/2167702614565359

Wiecki, T. V., Sofer, I., & Frank, M. J. (2013). HDDM: Hierarchical Bayesian estimation of the drift-diffusion model in Python. *Frontiers in Neuroinformatics*, *7*, 14. https://doi.org/10.3389/fninf.2013.00014

Williams, D. R., Mulder, J., Rouder, J. N., & Rast, P. (2021). Beneath the surface: Unearthing within-person variability and mean relations with Bayesian mixed models. *Psychological Methods*, *26*(1), 74–95. https://doi.org/10.1037/met0000270

Wood, S. N. (2017). *Generalized additive models: An introduction with R* (2nd ed.). Chapman; Hall/CRC. https://doi.org/10.1201/9781315370279